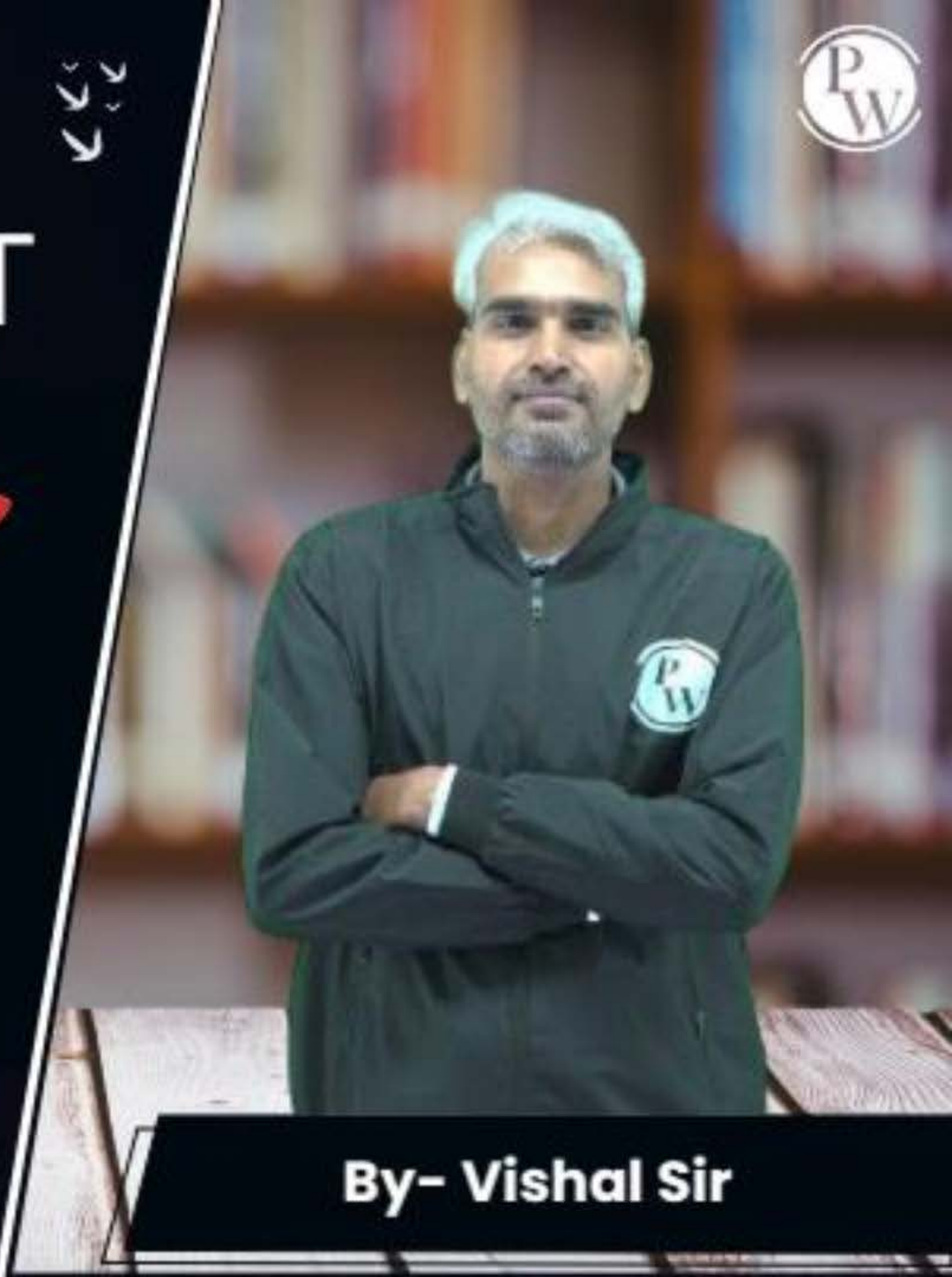
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 10



By- Vishal Sir

Recap of Previous Lecture



- Topic: Transitive closure
- Topic: Complement of a relation
- Topic: Inverse of a relation
- Topic: Composite of two relations

Slide

Topics to be Covered



- Topic — Equivalence relation
- Topic — Equivalence class
- Topic — Partition of a set
- Topic — Number of partitions of a set

Slide

H.W.

Q. Let X and Y are two sets such that

$|X^c| = 15$

$|Y^c| = 8$

and $|X \cap Y^c| = 5$

then $|X^c \cap Y| = ?$

H.W.

Q. Let X and Y are two sets such that

$|X^c| = 15$

$|Y^c| = 8$

and $|X \cap Y^c| = 5$

then $|X^c \cap Y| = ?$

$|X^c| = c + d = 15 - ①$

$|Y^c| = a + d = 8 - ②$

$|X \cap Y^c| = a = 5$

$|X^c \cap Y| = c = ?$

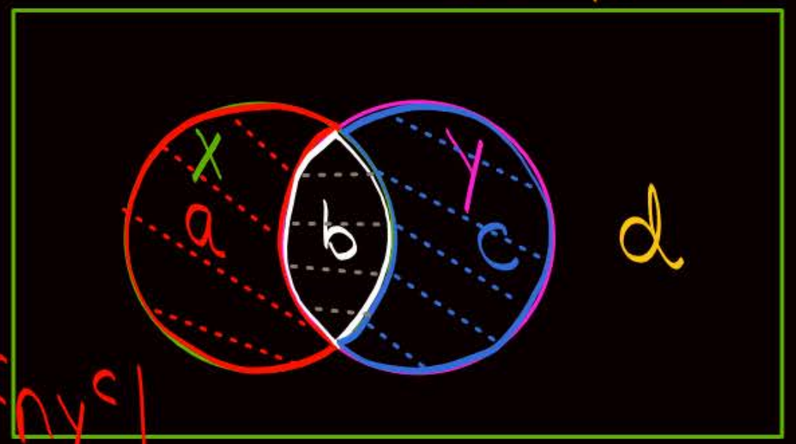
Eq① − Eq② ⇒ $c - a = 7$

$c - 5 = 7$

$c = 12$

$c + d = 15$

$d = 3$



$(a + c + d) = 5 + 12 + 3 = 20$

$|X^c \cup Y^c| = |X^c| + |Y^c| - \underbrace{|X^c \cap Y^c|}_{d}$

$= 15 + 8 - 3$

$= 20$ Ans

H.W

Q: The expression $(A \cup B \cup C') \cap ((A' \cap B' \cap C) \cup C')$ is equivalent to:

(a) $(A \cup B) - C'$

(b) $(A \cup B) \cap C$

(c) $(A \cup B) \cap C'$

(d) None of the above

$\{(A \cup B) \cup C'\} \cap \{((A' \cap B') \cap C) \cup C')\}$

$\{(A \cup B) \cup C'\} \cap \{\{(A \cap B') \cup C'\} \cap \{C \cup C'\}\}$

$\{\overline{(A \cup B)} \cup C'\} \cap \{\overline{(A' \cap B')} \cup C'\}$

$\overline{(x \cup p)} \cap \overline{(y \cup p)} = (x \cap y) \cup p$

$\{((A \cup B) \cap \overline{(A' \cap B')}) \cup C'\}$

$\overline{(A \cup B)}' = (A \cup B)$

$\emptyset \cup C' = C'$

H.W

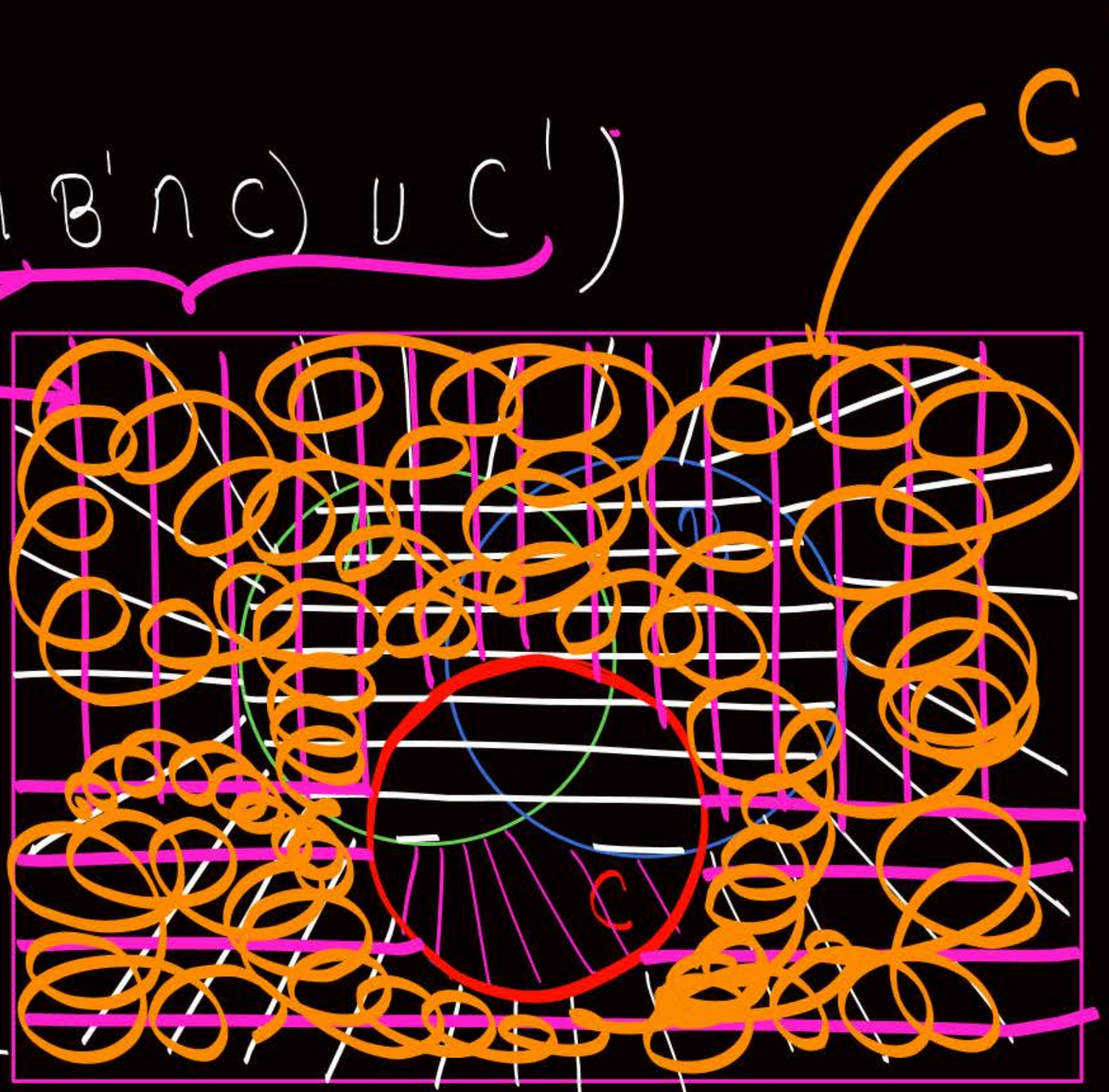
Q: The expression $(A \cup B \cup C') \cap ((A' \cap B' \cap C) \cup C')$ is equivalent to:

(a) $(A \cup B) - C'$

(b) $(A \cup B) \cap C$

(c) $(A \cup B) \cap C'$

(d) None of the above



H.W.

MSQ: Let 'P' be the set of all human beings, and R is the relation defined on set 'P' by the rule $(a, b) \in R$ if and only if blood group of a is same as blood group of b

then Relation R is

(a) Reflexive

(b) Symmetric –

(c) Transitive –

(d) Anti-symmetric –

let $P_1 \neq P_2$

if blood group of P_1 is same as P_2 then blood group of P_2 is same as P_1

i.e. if $(P_1, P_2) \in R$ then $(P_2, P_1) \in 2$

∴ Symmetric

H.W MSQ: Let 'P' be the set of all human beings, and R is the relation defined on set 'P' by the rule
$(a, b) \in R$ if and only if 'a' is brother of 'b'

then Relation R is

* ~~(a) Reflexive~~ {Actually relation R is ir-reflexive relation}
  if (Ram, Geeta) $\in R$, then (Geeta, Ram) $\notin R$
  (Ram: male, Geeta: female)
  But, if (Ram, Shyam) $\in R$, then (Shyam, Ram) $\in R$
  (Ram: male, Shyam: male)
* ~~(b) Symmetric~~
* ~~(c) Transitive~~
* ~~(d) Anti-symmetric~~
* (e) None of the above

if (Ram, Mohan) $\in R$ & (Mohan, Seeta) $\in R$

then (Ram, Seeta) $\in R$

Equivalence Relation



Topic : Equivalence relation

A relation R on set A is called an Equivalence relation if and only if relation R is

1. Reflexive
and 2. Symmetric
and 3. Transitive

Equivalence Relation = Reflexive + Symmetric + Transitive

Slide

eg. Let $A = \{1, 2, 3\}$

① $\Delta_A = R_1 = \{(1,1), (2,2), (3,3)\}$

Reflexive ✓
Symmetric ✓
Transitive ✓
∴ Equivalence Rel^n

Δ_A is the smallest Equivalence relation on set A

② $R_2 =$ "$A \times A$"

Reflexive ✓
Symmetric ✓
Transitive ✓
∴ Equivalence Relation

'$A \times A$' is the largest Equivalence relation on set A

Which of the following are Equivalence rel^n on set A

eg. Let $A = \{1, 2, 3\}$

$R_3 = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}$

Reflexive ✓
Symmetric ✓
Transitive ✓

$\therefore$ Equivalence Rel^n

$R_4 = \{(1,1) (2,2), (3,3), (1,3), (3,1)\}$

Reflexive ✓
Symmetric ✓
Transitive ✓

$\therefore$ Equivalence Rel^n

$R_5 = \{(1,1) (2,2), (3,3), (2,3), (3,2)\}$

Reflexive ✓
Symmetric ✓
Transitive ✓

$\therefore$ Equivalence Rel^n

eg. Let $A = \{1, 2, 3\}$

$R_6 = \{(1,1), (2,2), (3,3), (1,2), (2,1), (1,3), (3,1)\}$

Reflexive ✓

Symmetric ✓

R_6 is an Equivalence rel^n or not ?

$(2,1) \in R_6$ & $(1,3) \in R_6$

$\therefore$ for transitivity

$(2,3)$ must be present in R_6

But $(2,3) \notin R_6$ $\therefore$ R_6 is not transitive

Hence, R_6 is not an Equivalence Rel^n

- Check whether the statement is true or false:

> " Every symmetric as well as transitive relation on set 'A', is also a reflexive relation on set A "

→ This statement is false

eg: ① let $A = \{1, 2, 3\}$
& $R_1 = \{\ \}$ ← Empty rel^n ← it is symmetric & transitive but not reflexive

② let $A = \{1, 2, 3\}$
$R_2 = \{(1,1), (2,2), (1,2), (2,1)\}$ ← it is symmetric, transitive but not reflexive



Topic : Equivalence Class

→ Let R is an Equivalenc relation on set A.

→ Equivalence class of an element $x \in A$ w.r.t. Equivalence relation R can be denoted by $[x]$ and it is defined as,

$$[x] = \{ y \mid x \overset{R}{} y \}$$

is a set — of elements y — s.t. — x relates with y in rel^n R.

$$[x] = \{ y \mid (x, y) \in R \}$$

Slide

eg: Let $A = \{1, 2, 3, 4, 5\}$
and R is an Equivalence relation on set A.

$R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (2,1), (4,5), (5,4)\}$

Find Equivalence class of every element of set A w.r.t. Equivalence relation R.

$[1] = \{1, 2\}$ $\therefore [1] = [2]$

$[2] = \{2, 1\} = \{1, 2\}$

$[3] = \{3\}$

$[4] = \{4, 5\}$

$[5] = \{5, 4\} = \{4, 5\}$

$\therefore [4] = [5]$

'3' distinct Equivalence classes for the elements of Set A.

- Note :→ (1) Equivalence class of two distinct elements of the set may be equal { eg. [1]=[2] }

Note :- (2) Set of all distinct (Not mandatory) Equivalence classes of elements of set A will be a partition of set A with respect to given Equivalence relation

eg: Let $A = \{1, 2, 3, 4, 5\}$
and R is an Equivalence relation on set A.

$R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (2,1), (4,5), (5,4)\}$

Find Equivalence class of every element of set A w.r.t. Equivalence relation R.

$[1] = \{1, 2\}$ $\therefore [1] = [2]$

$[2] = \{2, 1\} = \{1, 2\}$

$[3] = \{3\}$

$[4] = \{4, 5\}$

$[5] = \{5, 4\} = \{4, 5\}$

$\therefore [4] = [5]$

'3' distinct Equivalence classes for the elements of Set A.

Set of all distinct Equivalence classes of elements of set A defines the partition of set A w.r.t. given Equivalence relation

$\therefore$ Partition of set $A = \{1, 2, 3, 4, 5\}$ w.r.t. Equivalence relation R is $= \{\{1, 2\}, \{3\}, \{4, 5\}\}$



Topic : Partition of a set

- Partition of a "set A" is the set of non-Empty subsets of set A such that each element of set A appeares in Exactly one non-Empty subset of that set.

(or)

- Let A is any set and $A_1, A_2, A_3, \ldots A_k$ are non-Empty subsets of set A; then

$\{A_1, A_2, A_3, \ldots \ldots A_k\}$ is a partition of set A if and only if (i) $A_i \cap A_j = \phi$, $\forall i \neq j$ and (ii) $\bigcup_{i=1} A_i = A$

Slide

eg:- Let $A = \{1, 2, 3, 4, 5\}$
Which of the following is/are Partitions of set A.

✗ (a) $\{\{1,2,3\}, \{3,4\}, \{5\}\}$ — $\cap \neq \emptyset$ ∴ Not a Partition of A

✗ (b) $\{\{1,2\}, \{3\}, \{4\}\}$ — $\{1,2\} \cup \{3\} \cup \{4\} = \{1,2,3,4\} \neq A$ ∴ Not a partition, '5' is not included in any subset.

✗ (c) $\{\{1,2,3\}, \{4\}, \{5\}, \{\}\}$ — Empty subset of Set A is not allowed in Partition.

✓ (d) $\{\{1,2\}, \{3\}, \{4\}, \{5\}\}$

✓ (e) $\{\{1,2,3,4,5\}\}$ is is partition of A w.r.t Equivalence Rel^n 'A×A'

✓ (f) $\{\{1,2\}, \{3\}, \{4,5\}\}$

* Find the number of partitions of "set A" when

(1) $|A| = 0$

(2) $|A| = 1$

(3) $|A| = 2$

(4) $|A| = 3$

(5) $|A| = 4$

① Let $A = \{\ \}$, i.e. $|A| = 0$

Partition of set A = { }

Partition of set A is a set such that it contains Non-Empty subsets of set A

But if 'A' itself is an Empty set, then there will be no non-empty subset of A, and hence no element will be present in the set w.r.t. Partition of set A

No element will be present.

Partition itself can be an Empty set, but Empty set can never be a member of a set corresponding to the Partition

i.e, if $|A| = 0$ then number of partitions of set A = 1

② Let $A = \{a\}$, i.e. $|A| = 1$

$|A| = 1 \Rightarrow \{1\}$ — Non-Zero, non-negative integer partition of '1'; in the Partition there will only '1' subset of size = 1

Partition of set $A = \{\{a\}\}$ — it is the only partition possible w.r.t. set $A = \{a\}$

Partition of set A is a set → such that elements will be non-empty subsets of set A

↳ w.r.t. set $A = \{a\}$, there is only '1' non-empty subset and it is $\{a\}$ itself

i.e, **If $|A| = 1$ then number of partitions of set $A = 1$**

(3) Let $A = \{a, b\}$ i.e. $|A| = 2$

$|A| = 2 \Rightarrow$ { $1 + 1$ — Two subsets each of size = 1 (or) 2 — only one subset of size = 2 }

Non-zero non-negative integer partition of 2

Partitions of set $A = \{\{a, b\}\}$ and $\{\{a\}, \{b\}\}$

$\{\{a, b\}\} = \{\{b, a\}\}$

$\{\{a\}, \{b\}\} = \{\{b\}, \{a\}\}$

in this partition there is only one subset of size = 2

In this partition there are two subsets each of size = 1

i.e, If $|A| = 2$, then number of partitions of set $A = 2$

(4) Let $A = \{a, b, c\}$ i.e. $|A| = 3$

$|A| = 3 \Rightarrow 1+1+1$ — 3 subsets each of size = 1

$1+2 = 2+1$ — Two subsets one of size = 1 & other of size = 2

3 — only one subset of size = 3

Partitions of set $A = \{\{a,b,c\}\}$ & $\{\{a\},\{b,c\}\}$ & $\{\{a\},\{b\},\{c\}\}$

&

$\{\{b\},\{a,c\}\}$

&

$\{\{c\},\{a,b\}\}$

i.e, **If $|A| = 3$, then Number of partitions of set $A = 5$**

|A| ⇒ 3 ⇒ 1+1+1

1+2

④ Let $A = \{a, b, c\}$ i.e. $|A| = 3$

Partitions of set A = $\{\{_\},\{_\},\{_\}\}$ or $\{\{_\},\{_,_\}\}$ or $\{\{_,_,_\}\}$ (3)

$\{a\}\ \{c\}\ \{b\}$
$\{a\},\{b,\}\ \{c\}$
$\{b\},\{a\}\ \{c\}$
$\{b\},\{c\}\ \{a\}$
$\{c\},\{a\}\ \{b\}$
$\{c\},\{b\}\ \{a\}$

$\frac{{}^3C_1 * {}^2C_1 * {}^1C_1}{3!}$ or ${}^3C_1 * {}^2C_2$ or ${}^3C_3$

$= 1$ $= 3 * 1 = 3$ $= 1$

i.e., If $|A| =$, then number of partitions of set A = $1 + 3 + 1 = 5$

$|A| = 1 \Rightarrow 1$

$|A| = 2 \Rightarrow 1 + 1$

2

$|A| = 3 \Rightarrow 1 + 1 + 1$

$1 + 2 = 2 + 1$

3

$|A| = 4 \Rightarrow 1 + 1 + 1 + 1$

$1 + 1 + 2$

$1 + 3$

$2 + 2$

4

$|A| = 5 \Rightarrow$

$1+1+1+1+1$

$1+1+1+2$

$1+1+3$

$1+2+2$

$1+4$

$2+3$

5

$|A| = 6 \Rightarrow$

$1+1+1+1+1+1$

$1+1+1+1+2$

$1+1+1+3$

$1+1+2+2$

$1+1+4$

$1+2+3$

$1+5$

$2+2+2$

$2+4$

$3+3$

6

⑤ Let $A = \{a, b, c, d\}$, i.e. $|A| = 4$, then find no. of Partitions of set A

$|A| = 4 \Rightarrow 1+1+1+1 = \dfrac{{}^4C_1 * {}^3C_1 * {}^2C_1 * {}^1C_1}{4!} = 1$

$1+1+2 = \dfrac{{}^4C_1 * {}^3C_1 * {}^2C_2}{2!} = 6$

$1+3 = {}^4C_1 * {}^3C_3 = 4$

$2+2 = \dfrac{{}^4C_2 * {}^2C_2}{2!} = 3$

$4 = {}^4C_4 = 1$

if $|A| = 4$ then number of Partitions of Set $A = 1+6+4+3+1 = 15$

H.W:

⑥ Let $A = \{a, b, c, d, e\}$ i.e. $|A| = 5$,
Find the total no. of Partitions possible of set A

$|A| = 5 \Rightarrow$ $1+1+1+1+1$

$1+1+1+2$

$1+1+3$

$1+2+2$

$1+4$

$2+3$

5

H.W.

$|A| = 6 \Rightarrow$

$1+1+1+1+1+1$

$1+1+1+1+2$

$1+1+1+3$

$1+1+2+2$

$1+1+4$

$1+2+3$

$1+5$

$2+2+2$

$2+4$

$3+3$

6

(7) Let $A = \{a, b, c, d, e, f\}$

i.e. $|A| = 6$,

Find total no. of partitions possible of set A.



2 mins Summary

- Topic: Equivalence relation
- Topic: Equivalence class
- Topic: Partition of a set
- Topic: Number of partitions of a set

THANK - YOU